

Chaotic Systems and Causal Inference through Convergent Cross Mapping

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Abstract—In this report, a two-variable discrete dynamical system, X and Y , is analyzed. In particular, it is shown how the Convergent Cross Mapping (CCM) method is able to determine the direction of causality between the two variables and the strength with which they influence each other.

I. INTRODUCTION

The system of equations considered is the following:

$$\begin{cases} X(t+1) = X(t) [r_X - r_X X(t) - \beta_{XY} Y(t)] \\ Y(t+1) = Y(t) [r_Y - r_Y Y(t) - \beta_{YX} X(t)] \end{cases} \quad (1)$$

where $r_X = 3.8$, $r_Y = 3.5$, $\beta_{XY} = 0.02$, $\beta_{YX} = 0.1$, $X_0 = 0.4$, and $Y_0 = 0.2$.

β_{XY} and β_{YX} are the parameters that describe the influence of Y on X and that of X on Y , respectively, while r_X and r_Y are indicators of the chaotic behavior of the system.

It is easy to understand that the two observables are not separable; rather, they are weakly dependent on one another. This means that the behavior of the two time series is somehow connected, although in a rather unclear way. This feature is crucial when choosing methods to analyze correlation and causality.

In the case of independent time series, the first evidence of a possible causal relationship between the series is their correlation. In most cases, performing a Cross-Correlation test provides a good starting point for further investigation. A peak in the correlation value between the two series at a given time lag can certainly be interpreted as a signal. Commonly, one then proceeds by performing a Granger Causality (GC) test. Conceptually:

$$X \text{ Granger-causes } Y \iff Y_{t+h} | (Y_{k \in \text{past}}, X_{k \in \text{past}}) \text{ predicts } Y_{t+h} \text{ better than } Y_{t+h} | (Y_{k \in \text{past}}) \quad (2)$$

The key requirement of GC is separability, namely that the information about a causative variable is specific and unique to that variable and can be removed by eliminating the variable from the model.

The non-separability of the system prevents the use of GC for the system under analysis. The tool chosen in this report to investigate possible causal links in the system is Convergent Cross Mapping (CCM), which involves a paradigm shift with respect to GC. In dynamical systems theory, the variables of a time series are causally linked if they originate from the same dynamical system, that is, if they share a common attractor manifold M . When a variable X is a causal factor for a variable Y , information about the states of X can be recovered from Y , but not vice versa. Convergent Cross Mapping tests

causality by measuring the extent to which the historical record of values of Y can reliably estimate the states of X . This happens only if X causally influences Y .

In practice, the two attractor manifolds M_X and M_Y are constructed, and one checks whether nearby points on the manifold M_Y can be used to identify nearby points on the manifold M_X . In this case, Y can be used to predict X . This, in a perhaps counterintuitive way, means that X causes Y , precisely because Y contains information about X .

II. ANALYSIS AND METHODS

When attempting to analyze the system through a “classical” approach, one realizes that it is inconclusive, or rather, that it does not even allow for a deeper analysis to be carried out.

Indeed, given the two time series under analysis:

$$\begin{aligned} X &= X(1), X(2), \dots, X(L) \\ Y &= Y(1), Y(2), \dots, Y(L) \end{aligned} \quad (3)$$

where L is the length of the time series, it is particularly difficult to infer a relationship of influence between the two. By performing a simple Pearson test, where r represents the correlation value and p the p-value, one obtains:

r	p
0.0054	0.865

These values seem to indicate an almost null correlation between the two time series. By analyzing the behavior of the two series in detail, one can observe that in some time intervals they are strongly correlated, in others they are practically uncorrelated, and in still others they are strongly negatively correlated (Figure 1, Panel A). The succession of these chaotic behaviors therefore leads to an overall correlation value that is practically zero.

Even when performing a Cross-Correlation test, the result does not change: the two series do not show any correlation peak for any specific time lag. The correlation value in fact oscillates markedly, without ever reaching a threshold that would allow one to claim the existence of any actual correlational link (Figure 1, Panel B).

At first glance, therefore, the correlation analysis between the two series does not seem to indicate the existence of a relationship, either instantaneous or delayed. However, the absence of correlation is not sufficient to state that there is no causal link.

It is therefore necessary to investigate further using more refined methods. The use of Convergent Cross Mapping falls within this framework.

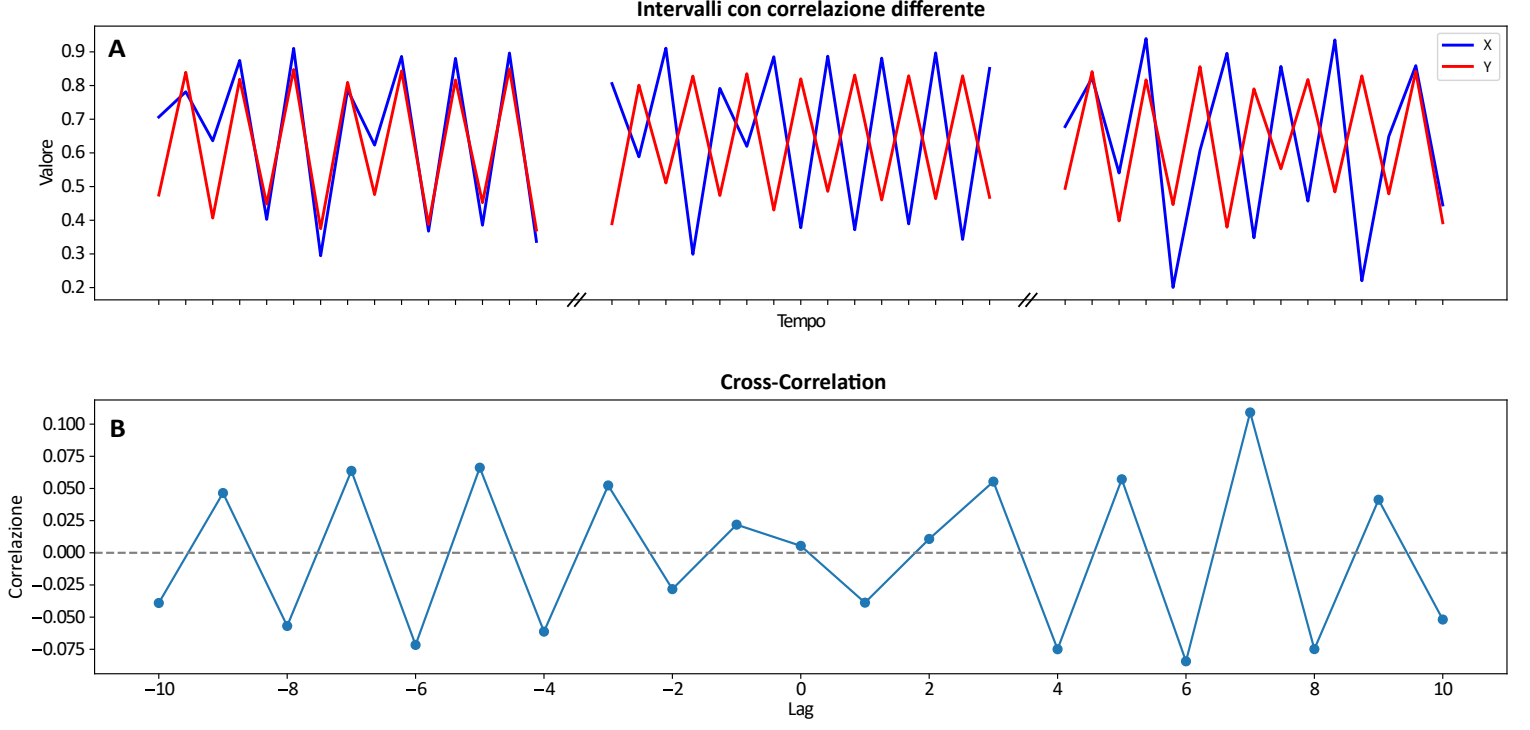


Figure 1. Panel A: three different intervals with three different correlation values. The first interval has $r > 0.9$, the second has $r < -0.9$, and the third has $-0.05 < r < 0.05$. Panel B: Cross-Correlation test; no specific time lag value shows a peak in the correlation value.

CCM involves computing the delay-coordinate vectors:

$$\begin{aligned}\vec{x} &= (X(t), X(t-\tau), \dots, X(t-(E-1)\tau)) \\ \vec{y} &= (Y(t), Y(t-\tau), \dots, Y(t-(E-1)\tau))\end{aligned}\quad (4)$$

where τ is the time lag and E is the embedding dimension. There are several algorithms for determining the optimal values of τ and E ; in this report, unless otherwise specified, $\tau = 1$ and $E = 2$.

The attractor manifold M_x is then defined as the set:

$$M_x = \{\vec{x}(t) \mid \forall t \in [1 + (E-1)\tau, L]\} \quad (5)$$

The same can be done for M_Y . Figure 2 shows what the two manifolds look like in the case $E = 3$. Increasing E increases the dimensionality of the manifold.

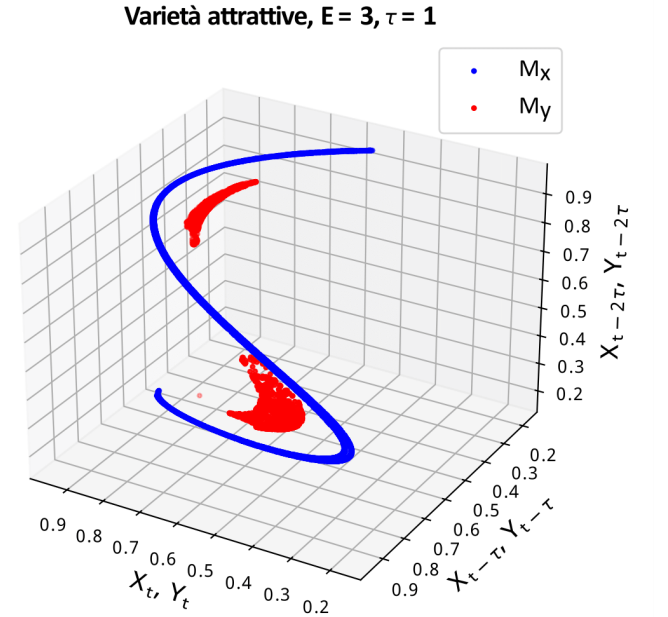


Figure 2. 3-D graphical illustration of the attractor manifolds M_x and M_y with 1000 points. The three axes represent the value of the series, X and Y , at a precise time instant: t , $t - \tau$, and $t - 2\tau$.

Given a certain t , one can identify $\vec{x}(t)$ on the manifold and find the $E + 1$ nearest points on that manifold. These points on the manifold correspond to certain time instants, which can be labeled as $t_1, t_2, \dots, t_{E+1}$. The time indices are then used

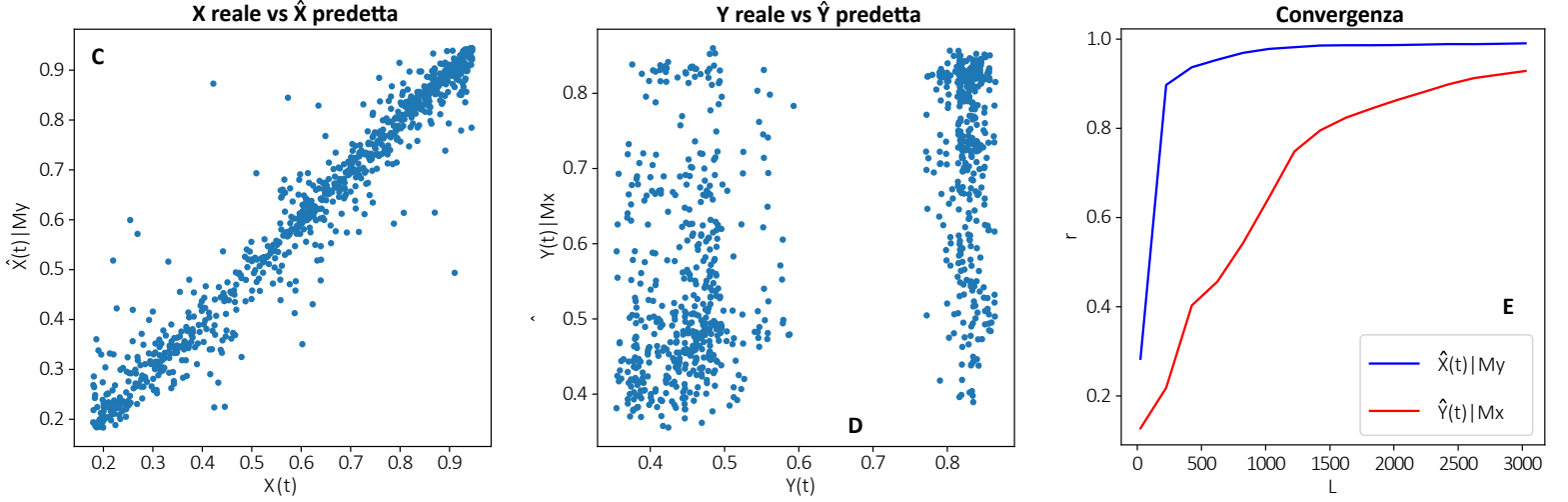


Figure 3. Panel C: scatter plot of the real values of X against the predicted values $\hat{X}$. The high level of correlation between the two series is evident. Panel D: scatter plot of the real values of Y against the predicted values $\hat{Y}$. Two clusters are visible, but the predictive model is not as accurate as in the other case. Panel E: convergence toward the maximum correlation value $r = 1$ for the two series $\hat{X}$ and $\hat{Y}$: the series $\hat{X}$ converges much faster.

to identify the corresponding points in Y .

We now define the model that predicts Y given M_x :

$$\begin{aligned} \hat{Y}|M_x &= \sum_{i=1}^{E+1} w_i Y(t_i) \\ w_i &= \frac{u_i}{\sum_{j=1}^{E+1} u_j} \quad \text{where } j = 1, \dots, E+1 \\ u_i &= \exp \left[-\frac{d(\mathbf{x}(t), \mathbf{x}(t_i))}{d(\mathbf{x}(t), \mathbf{x}(t_1))} \right] \end{aligned} \quad (6)$$

At this point, the correlation coefficients between Y and $\hat{Y}|M_x$ can be computed. A significant correlation implies that some information about Y is preserved in X , and therefore one can state that Y causes X . In order to confidently assert that Y causes X , it is also necessary to verify convergence, namely that as the library size L increases, the prediction of the values should improve, approaching the maximum correlation value of 1.

Clearly, the same reasoning applies in the opposite direction when considering M_y and $\hat{X}|M_y$.

In the system under examination, the attractor manifold M_y generates a series $\hat{X}$ that is very similar to the real series X , meaning that X shows a high level of predictability given M_y (Figure 3, Panel C). Conversely, the predicted $\hat{Y}$ differs from the real Y (Figure 3, Panel D).

The computed correlation values, with $L = 1000$, are:

$r(X \hat{X}M_y)$	$r(Y \hat{Y}M_x)$
0.977	0.629

Both correlation values tend to improve as the library size L increases. This is clearly visible in Figure 3, Panel E, where the values of $r(X|\hat{X}M_y)$ and $r(Y|\hat{Y}M_x)$ were computed for $25 < L < 3100$. The notion of convergence, $r \xrightarrow{L \rightarrow \infty} 1$, is necessary in order to claim causality between the series.

The fact that the correlation value between X and $\hat{X}$ is so high, together with its particularly fast convergence, leads to

the conclusion that X causes Y . Nevertheless, given the lower but still significant correlation value, one can also state that Y causes X . However, it should be specified that the strength of this causal direction is considerably weaker, also considering that convergence is much slower.

III. CONCLUSIONS

The chaotic system analyzed has shown that pure correlation analysis between variables can be of limited relevance when the goal is to investigate causality. Indeed, it was observed that the trends of the variables may be more or less similar across different time intervals, making it particularly difficult to infer the existence of a connection.

The Convergent Cross Mapping method, through the computation of r and convergence, proved to be extremely effective in demonstrating that there is indeed a strong bidirectional cause-effect relationship between the two variables, especially from X toward Y , rather than from Y toward X .

IV. REFERENCES

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